INCH POUND

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# COMMERCIAL ITEM DESCRIPTION

## TRACTOR, AIRCRAFT TOWING, FLIGHT LINE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the requirements for one type of Flight Line Tow Tractor (FLTT), hereafter referred to as tow tractor. The tow tractor is based on a commercial truck chassis, also known as a chassis cab, which is modified for use as a tow tractor in worldwide service. These chassis are generally sold to final stage manufacturers who complete the vehicle. The tow tractor is capable of towing trailers up to 40,000 pounds, and military aircraft up to 70,000 pounds on grades up to two and one half percent.

2. SALIENT CHARACTERISTICS. Failure of the tow tractor to meet the salient characteristics of this CID shall be cause for rejection of the tow tractor. The tow tractor shall have a life expectancy of 15 years, and shall be free from faulty construction and any defect that affects appearance, serviceability, and durability. The tow tractor shall be representative of the quality and workmanship found in tow tractors offered for new commercial sale.

2.1 Gross vehicle weight (GVW). The tow tractor shall have a minimum GVW of 8000 pounds. However, the GVW shall not exceed gross vehicle weight rating (GVWR) less l500 pounds. The minimum GVWR required for this application is 9500 pounds.

2.2 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length: 200” without couplers
Width: 100”
Height: 100”

2.3 Drawbar pull (DBP). The tow tractor shall have a minimum DBP of 4,500 pounds on dry level concrete.

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LGEC, 480 Richard Way Blvd. Suite 200, Robins AFB, GA 31098-1640 or emailed to bob.yohe@robins.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil

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2.4 Power train. The tow tractor shall have a diesel engine of not less than six cylinders, automatic transmission, and power train sufficient to develop the required drawbar pull and obtain the required tow tractor speeds. Power train components shall have adequate torque and power ratings for the imposed torque and power. The tow tractor power train shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of +125 degrees F without exceeding the manufacturer's maximum allowable temperatures on the individual power train components. The diesel engine shall operate satisfactorily when using Jet A-1 (JP-8) fuel. Power train shall feature a standard (type C) winterization package that is good for starting down to minus 20F, or when specified (see 6.2 c), a heavy-duty (type A) winterization package good for storage down to minus 65F and starting to minus 40F. Winterization package components with external power requirements shall be compatible with 110VAC/20A circuitry. The tow tractor shall meet applicable 50-state exhaust and evaporative emissions standards in effect at the time of vehicle manufacture. The tow tractor shall include an anti-restart, as well as a neutral safety start feature. The rear axle shall have a limited-slip differential and a reduction ratio equal to or greater than 4.1:1. The rear axle shall feature single rear wheels.

2.5 Driveline and steering. When specified (see 6.2 f), the tow tractor shall be offered in a part-time, four-wheel drive variant. The rear axle reduction ratio shall be the same on the two-wheel drive and the four-wheel drive tow tractor variants. Non-directional mud and snow tires are required on the four-wheel drive variant. The tow tractor shall have a maximum wall-to-wall turning radius of 21 feet or less. The wheelbase shall be between 90 and 100 inches. The balance of these two parameters shall be optimized.

2.6 Electro-magnetic interference (EMI). The tow tractor shall comply with EMI requirements of MIL-STD-461E for Air Force ground equipment.

2.7 Mobility. The tow tractor shall be capable of 35 MPH (minimum) on flat asphalt or concrete surfaces without a trailing load. The tow tractor shall be governed not to exceed 40 MPH (plus 0, minus 3), unless the rear axle reduction ratio prevents speeds greater than 40 MPH. The preferred method to control the tow tractor speed is through the selection of the axle gear ratio. Reverse speed shall be a minimum of 2.5 MPH on flat surfaces. The tow tractor shall be capable of towing trailers from a stop to 10 MPH (minimum) and aircraft from a stop to 5 MPH (minimum) on grades up to 2.5% on asphalt or concrete surfaces. The tow tractor shall be capable of operating on 10 percent side slopes and 12 percent grades without fluid spillage.

2.8 Suspension. Reserved.

2.9 Wheels and tires. The tow tractor shall be equipped with steel disc wheels and tubeless steel belted radial tires with an appropriate tread pattern. The wheels and tires shall be in accordance with the Tire and Rim Association recommendations for this application. All wheels and tires shall be uniform. Tire inflation pressures shall be noted on all wheel well openings. If tire loads and tire inflation pressures that exceed the specifications on the tire sidewall markings are required, a copy of the application approval from the tire manufacturer shall be added in the Technical Order.



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2.10 Braking system. The tow tractor shall feature a braking system in compliance with Federal Motor Vehicle Safety Standards (FMVSS) for vehicles of this GVWR.

2.11 Parking brake. The tow tractor shall be equipped with a parking brake that shall be in compliance with SAE J293.

2.12 Air system. When specified (see 6.2 d), the tow tractor shall have an air system for towing trailers with air brakes. The air system shall be an over-the-road commercial truck/tractor design to include a reservoir tank with heated and automatic drain valves, automotive compressor, dryer, trailer protection valve, low pressure indicator, etc. The glad hands shall be in accordance with SAE J318. The configuration of the air system shall be the same on the two-wheel drive and the four-wheel drive tow tractor variants.

2.13 Electrical system. The tow tractor shall be equipped with a 12V electrical system. The battery(ies) shall be maintenance free.

2.14 Frame structure. The commercial truck chassis shall be modified to the specified wheelbase by relocating the rear axle, suspension, and other affected components forward on the original chassis frame rails. Back-halved chassis or spliced-in rear sub-frames, creating a joint in the center of the chassis length, shall not be used in the modification. Chassis components shall be relocated and/or modified as necessary to avoid excessive loading, suboptimal drive angles, and driveline vibrations. Modifications to chassis components, especially load bearing suspension components, shall be performed using an engineering approach and with quality workmanship so as to avoid weakening the structural integrity of the component. Excess length at the rear of the chassis frame rails from the forward relocation of chassis components shall be removed using a process that leaves a smooth cut finish. Raw metal exposed by the modifications shall be primed and painted chassis color. The relocation of chassis components shall not result in improperly loaded fasteners.

2.15 Cab design. The tow tractor shall be equipped with a three-man cab. The first entrance step to the cab shall be a running board style step and shall not exceed 18 inches in height from the ground. Cab instrumentation shall feature an hour meter in addition to the standard instrumentation. When an air system for towing trailers is specified (see 6.2 d), cab instrumentation shall include an air pressure gauge and a low pressure indicator with buzzer and/or warning light. The tow tractor shall have a covered toolbox with a hinged lid located behind the cab in an easily accessible location. When specified (see 6.2 h), the tow tractor shall have an Air Conditioning system. The tow tractor cab shall be equipped with wipers, driver and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The cab shall feature a rear sliding window and a 12V power outlet (cigarette lighter type).



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2.16 Lighting. The tow tractor lighting shall be equivalent to the standard commercial offering to include a rear coupling light, parking lights, head lights (high and low beam), brake lights, turn signals, and hazard lights. Basic lighting system shall be in compliance with FMVSS requirements. A flashing amber beacon light shall be mounted above the roofline on the tow tractor's centerline. Two double-faced signal lamps (SAE J588) shall be mounted above the roofline, approximately 48 inches apart and centered on the tow tractor's centerline, and shall be integrated with the tow tractor's taillights. An adjustable aim floodlight, of not less than 3000 candlepower, shall be mounted on the tow tractor so that the rear working area is illuminated. When specified (see 6.2 g), the tow tractor shall be equipped with an adjustable spotlight on the driver's side. Spotlight shall be steerable from the driver's side inside the cab. The flashing beacon light, double-faced signal lamps, and adjustable flood light shall be mounted on a light bar that is not fastened to the cab that runs approximately one inch above the roofline.

2.17 Pintle hooks. The tow tractor shall be equipped with four pintle hooks that incorporate a cotter pin as a secondary-locking device, and conforming to CID A-A-52550, Type 1, Class 2 (old MS51335-2). Pintle hooks shall be attached to the pintle hook mounting structure with a minimum of four ½-13 or ½-20, Grade 8 fasteners.

2.17.1 Front pintle hook position. The front of the tow tractor shall be equipped with one center-mounted pintle hook capable of withstanding a minimum of 18,000 pounds DBP, with the coupling position of the pintle hook located at 18 ½ inches, plus or minus ½ inch, from the ground. Additionally, the front pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

a. The rated load shall be the load combination of forces that the pintle hook assembly and structure must support or resist in a static state.
b. The dynamic load shall be the rated load with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.
c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, which ever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

2.17.2 Rear-outboard pintle hook positions. The rear of the tow tractor shall be equipped with two outboard pintle hooks capable of withstanding a minimum of 6,000 pounds DBP, located in the same vertical plane, with the pintle hook centerlines at 43 ½ inches, plus or minus ½ inch, to each side of the tow tractor centerline, with the coupling positions of the pintle hooks located at 18 ½ inches, plus or minus ½ inch, from the ground. The rear-outboard pintle hook assemblies and mounting structures shall be designed to accommodate the greatest force exerted by the drawbar with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.



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2.17.3 Rear-center pintle hook position. The rear of the tow tractor shall be equipped with one center-mounted, extendable pintle hook capable of withstanding a minimum of 6,000 pounds DBP in the extended position, a minimum of 18,000 pounds DBP in the retracted (closed) position, and with pintle hook coupling positions that are height adjustable from 12 inches to 18 ½ inches from the ground. A minimum of four evenly spaced or coupling positions (including 12 inches and 18 ½ inches) shall be provided. Height adjustment shall be easily made by the operator without the use of tools. The rear-center extendable pintle hook shall not extend more than 10 ½ inches beyond the rear plane of the truck when retracted for towing. The extendable unit shall consist of a self-locking sliding bar that shall allow the hook backward movement of not less than 10 ½ inches, or more than 12 inches, and a lateral swing movement of 45 degrees in either direction from the centerline of the tow tractor. The unit shall be fitted with rollers, and provisions shall be made for lubricating all moving parts. The normal towing position shall be with the pintle hook/hitch assembly in the retracted and straight positions, rather than the extended and swung positions. The extendable bar shall be equipped with a safety grab handle, attached to the pintle hook/hitch assembly, and positioned above the pintle hook, to allow for easy user access and function of the extendable hitch's front-to-rear and side-to-side hitch motion, as well as the pintle hook height adjustments. The extendable bar shall be easily removable to facilitate inspection and maintenance. The extendable/swing features are required to facilitate connection of the pintle hook to the trailer by a single operator, rather than a method requiring two users. Once the lunette or towing eye of the towed load is secured, the extendable unit shall be engaged in the closed and straight positions for towing by the backward movement of the tow tractor until the self-locking device is secure. Any shear pins, locking pins, or components that are designed to be removed for adjustments by the user shall be attached to the tow tractor with lanyards. Those shear pins that use a locking pin for retention shall utilize a locking pin that is a double-loop hair-pin design. If constructed of mild steel, the hitch shall have hardened steel bushings and pins at all pivot and locking points. The bushings shall be easily removed for service. Additionally, the rear-center pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

a. The rated load shall be the load combination of forces that the pintle hook assembly and structure must support or resist in a static state.
b. The dynamic load shall be the rated load with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.
c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, which ever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

2.18 Breakaway cable hookup. The tow tractor shall be equipped with a rear trailer emergency breakaway cable hookup, capable of mating with NAS 1281C10 spring snap hook. The breakaway chain attachment shall be installed on the bumper or frame, on the tow tractor's centerline, within 8 inches (vertical distance) above the center pintle hook while in its uppermost height position, and within 4 inches forward of the base of the pintle hook. It shall not be installed on the pintle hook or pintle hook assemblies.



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2.19 Safety chain attachment points. The tow tractor shall have two safety chain attachment points, one located on each side of pintle hook, symmetrical about the pintle hook, and located at a distance of 26 inches in height from the ground for use during towing on the center, rear pintle hook only. The vertical centerline of the safety chain attachment points shall be located at a parallel distance of not greater than 6 inches to the left and right of the tow tractor's centerline, and shall in no way be a part of or attached to the pintle hook. Additionally, the rear edge of the safety chain attachment points shall be within 2 inches forward of the base of the pintle hook. Each of the attachment points shall be capable of withstanding a maximum shock load of 5,000 pounds and shall have the ability for standard safety chain hooks to be attached. The inside radius of the safety chain hook is between one inch and three inches.

2.20 Trailer light harness receptacle. The tow tractor shall be equipped with an SAE J560b electrical receptacle, provided on the tow tractor's centerline, within 12 inches (vertical distance) above the center pintle hook while in its uppermost height position. All terminals of the receptacle shall be wired in accordance with SAE J560b. The tow tractor shall be equipped with an aftermarket, heavy-duty trailer-light circuit protection system that shall isolate the trailer's lighting circuits from the tow tractor's electrical system. This system shall power any trailer lights directly from the tow tractor's battery, and be capable of handling up to 20 amps.

2.21 Fire extinguisher. The tow tractor shall be equipped with two 5 pound (minimum) dry chemical type fire extinguishers with mounting brackets, and a ULC rating of 4A:20B:C (minimum). Extinguishers shall be mounted on the left and right side of the tow tractor, rear of the cab.

2.22 Finish. As specified (see 6.2 e), the tow tractor shall be painted Dark Green (Color 24052, FED-STD-595) or Desert Sand (Color 23448, Fed-STD-595) using offeror's standard commercial paint system.

2.23 Corrosion protection. The tow tractor chassis and cab shall be rust proofed and undercoated in accordance with FED-STD-297. The level of corrosion treatment shall be tropical, and the use of proprietary material shall require Government approval.

2.24 Visibility. The tow tractor shall be so constructed as to provide maximum visibility in all directions. The three rear pintle hooks shall be visible either directly or via mirrors from the driver's position. If required, one mirror per pintle hook shall be provided to complete this visibility requirement. A plane defined by a line 18 inches above the pintle hooks and a point represented by the driver's eye shall not be obstructed by any structure or feature on the truck. The light bar is exempt from this requirement.

2.25 Air transportability. The tow tractor shall be air transportable on C-130, C-141, C-17 and C-5 aircraft without removing ballast or shoring to load the aircraft (referred to as approach shoring and step-up shoring). Use MIL-HDBK-1791 as a guide. In all transport configurations, the tow tractor shall be capable of being restrained and withstanding 2.0 G up, 4.5 G down, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The tow tractor shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.



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2.25.1 Air transportable configuration. The air transportable configuration shall include all equipment, ¾ tank of fuel, and no crewmembers. At least 6.0 inches sidewall and 6.0 inches overhead clearance shall be maintained between the tow tractor and the aircraft at all times during loading and flight. The restrained tow tractor shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

2.25.2 Equipment removal/reconfiguration. Preparation and restoration of the tow tractor for air transportability shall take no more than 15 minutes for two persons using common non-powered hand tools. All equipment removed shall be stored on the tow tractor; caps and plugs shall permit driving and storage in transport configuration.

2.25.3 Tie down points. The tow tractor shall be symmetrically restrained during air transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with the appropriate devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the tow tractor.

2.25.4 Shoring. The use of shoring during flight (referred to as load shoring, parking shoring, and sleeper shoring) is permitted, but not desired. However, the use of shoring is not an alternative to good tow tractor design.

2.26 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information.

1. Side and rear silhouette views of the tow tractor
2. Horizontal and vertical location of the center of gravity of the tow tractor in air transportable configuration, marked on the silhouette views
3. Shipping weight
4. Loading cubage
5. Overall height, width, and length
6. Front and rear axle loads
7. Tie down information

2.27 Identification plate. The tow tractor shall have an identification plate and a rear license plate mounting bracket. The identification plate shall include as a minimum the following.

1. Vehicle Nomenclature
2. Manufacturer's Model Number
3. Vehicle Identification (serial) Number or VIN
4. Manufacturer's address and phone number
5. Date of Manufacture
6. National Stock Number
7. Contract Number
8. Registration Number

2.28 Lubrication data plate. The tow tractor shall be provided with a lubrication data plate that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.



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2.29 Miscellaneous specifications. The tow tractor shall comply with SAE specifications ARP1247 and AIR 1316. Modifications to the truck chassis shall not result in exposed fuel supply or fill lines.

2.30 Technical Order (TO) requirements. The contractor shall provide copies of the tow tractor manuals for review 30 days prior to the time of tests or demonstration of the first tow tractor. The contractor's parts manuals shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. The contractor shall also provide with each tow tractor, a product familiarization videotape that verbally and visually provides all information for the operation and routine maintenance of the tow tractor and its components, using the commercial manuals as a baseline.

3. RECLAIMED MATERIAL. The contractor shall utilize reclaimed materials to the maximum extent practical. Used, rebuilt, or remanufactured components shall not be incorporated in the tow tractor.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance and configuration control. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be substantially the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance. The configuration of the offeror's added equipment or features shall be uniform on both the two-wheel drive and four-wheel drive variant unless otherwise approved by the contracting officer. The only differences between the two-wheel drive and four-wheel drive tow tractor variants shall be the chassis differences as delivered by the chassis cab manufacturer. The configuration of the options for an air-system for trailer towing and Air Conditioning (see 6.2 d and 6.2 h) shall be uniform when optioned separately or in combination

4.2 Commercial item requirement. The tow tractor furnished must meet the "commercial item" definition, as specified in FAR 2.101, as of the date of award. The offeror/contractor shall provide two copies of their commercial, descriptive catalogs with their offer, as specified in Clause I106. The offeror/contractor shall identify all modifications made to their commercial model in order to meet the performance and descriptive requirements of the CID or the referenced documents. In regard to the offered item, the offeror/contractor shall identify any enhancements or improvements to the performance requirements of paragraph 2 of the CID or the referenced documents.

4.3 Verification. Prior to delivery of the first tow tractor of each configuration (two-wheel drive and four-wheel drive), the contractor shall demonstrate or test at his facility, each requirement of the CID or referenced documents.

4.3.1 Gross vehicle weight (GVW). The first production tow tractor shall be weighed for verification of the minimum GVW.



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4.3.2 Overall dimensions. The first production tow tractor shall be measured for verification of the overall dimensions.

4.3.3 Drawbar pull (DBP). The first production tow tractor shall be tested for verification of the DBP on dry level concrete using a load sensing device and a fixed anchoring position. The drawbar position shall be horizontal and level.

4.3.4 Power train. The first production tow tractor shall be inspected for verification of the specified power train components. A copy of the manufacturer's signed application for approval shall be submitted to the government representative at first production verification with the vehicle documents for the following components: engine, transmission/transfer case, and axles, to include torque and power ratings, use of Jet A-1 or JP-8 fuels, emissions compliance, and engine shutdown system. The cooling system shall be certified by the vehicle manufacturer to satisfy all operating conditions at all ambient temperatures up to the specified, for both the engine and transmission. The winterization packages shall be certified by the vehicle manufacturer to satisfy the winterization conditions specified.

4.3.5 Driveline and steering. The first production tow tractor shall be demonstrated with a test drive by the government representative. The tow tractor wheelbase shall be measured for verification of the specified dimension. The tow tractor shall be tested for verification of the maximum wall-to-wall turning radius while in four-wheel coordinated steering mode. The fully ballasted first production tow tractor shall be tested as follows to demonstrate compliance with the turning radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the vehicle being tested.

b. Equipment required.

    1. A device suitable for measuring three times the length of the vehicle being tested with an accuracy of at least ±1-inch.

    2. Markers or marking device suitable for marking the pavement.

    3. A plumb bob suitable for locating a point on the pavement directly below a fixed point on the vehicle.

    4. A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.

c. Test Conditions.

    1. The tow tractor shall be fully ballasted and in operational form.


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2. The tow tractor steering system shall be fully operational, and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.

d. Test Procedure.

1. The vehicle shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.

2. Continue driving the slow full cramp circle.

3. At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the vehicle using the service brakes.

4. At each stop, place a plumb bob against the outermost point of the vehicle and mark the spot on the ground directly below where the plumb bob comes to rest. Measure and record the straight line distances between each pair of points, i.e., Lengths AB, BC, and CA.

5. Measure and record the straight line distances between each pair of points, i.e., Lengths AB, BC, and CA.

6. Calculate the wall-to-wall turning radius (R) as follows:

$R = D / 2$ where

$D = (AB \times BC \times CA) / (2 \times \sqrt{(S \times (S-AB) \times (S-BC) \times (S-CA))})$ and

$S = (AB + BC + CA) / 2$

7. Repeat steps (1) through (6) with the vehicle moving in the opposite direction.

4.3.6 Electro-magnetic interference (EMI). The first production tow tractor shall be tested by the contractor for verification of the specified EMI requirements prior to first production verification of the tow tractor. The tow tractor shall be tested for Radiated Emission per RE102 of MIL-STD-461E and Radiated Susceptibility per RS103 of MIL-STD-461E. The tow tractor shall be tested for Conducted Susceptibility per CS114, CS115, and CS116 of MIL-STD-461E. Test results shall be submitted to the government representative at first production verification.

4.3.7 Mobility. The first production tow tractor shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the tow tractor’s speedometer as the speed measuring device. The tow tractor shall be tested for verification of its capability to tow a 40,000 pound trailer on a 2.5 % grade at a minimum of 10 MPH on an asphalt or concrete surface, using the tow tractor’s speedometer as the speed measuring device. The tow tractor shall be tested for verification of its capability to operate on 10 percent side slopes and 12 percent grades without fluid spillage.



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4.3.8 Suspension. Reserved.

4.3.9 Wheels and tires. The first production tow tractor shall be inspected for verification of the specified wheels and tires. Documentation of the proper wheel and tire selection and deviations in tire pressures (if any) with references to the Tire and Rim Association shall be submitted to the government representative at the first production verification.

4.3.10 Braking system. The first production tow tractor shall be inspected for verification of an FMVSS compliant braking system.

4.3.11 Parking brake. The first production tow tractor shall be inspected for verification of the parking brake. Documentation of compliance with SAE J293 shall be submitted to the government representative at the first production verification.

4.3.12 Air system. A schematic of the air system plumbing shall be submitted to the government representative at first production verification.

4.3.13 Electrical system. The first production tow tractor shall be inspected for verification of the specified electrical system.

4.3.14 Frame structure. The first production tow tractor shall be inspected for verification of the chassis modifications as specified.

4.3.15 Cab design. The first production tow tractor shall be inspected for verification of the three-man cab, the entrance step and height, cab instrumentation, and specified equipment.

4.3.16 Lighting. The first production tow tractor shall be inspected for verification of the specified lighting and location. Documentation of the double-faced signal lamp's adherence to SAE J588 and the candlepower of the adjustable aim floodlight shall be submitted to the government representative at first production verification.

4.3.17 Pintle hooks. The first production tow tractor shall be inspected for verification of the specified pintle hooks and locations. A strength analysis report shall be submitted to the government representative at first production verification that contains supporting calculations to show that the front and rear pintle hook assemblies and mounting structures are in compliance with their respective strength specifications. The analysis shall include methodology leading to the arrival of values for static and dynamic loading, and ultimately the design load. Additionally, the front-center and rear-center pintle hook assembly and mounting structure calculations shall be supplemented with a Finite Element Analysis (FEA) to further support compliance with the strength specification.

4.3.18 Breakaway cable hookup. The first production tow tractor shall be inspected for verification of the specified breakaway cable hookup and location.



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4.3.19 Safety chain attachment points. The first production tow tractor shall be inspected for verification of the specified rear safety chain attachment points and locations. Documentation of the strength of the safety chain attachment points and mounting (through ratings and/or calculations) shall be submitted to the government representative at first production verification.

4.3.20 Trailer light harness receptacle. The first production tow tractor shall be inspected for verification of the specified J560b electrical receptacle and location. An electrical schematic of the receptacle and its circuitry shall be submitted to the government representative at first production verification.

4.3.21 Fire extinguisher. The first production tow tractor shall be inspected for verification of the specified fire extinguishers and mounting brackets.

4.3.22 Finish. Color sample cards with the specified colors shall be submitted at the first production verification.

4.3.23 Corrosion protection. The first production tow tractor shall be inspected for verification of the specified rust proofing and undercoating. Documentation of the application process and certification of the level of corrosion treatment shall be submitted to the government representative at the first production verification.

4.3.24 Visibility. The first production tow tractor shall be inspected for verification of the specified visibility. Measurements of the defined visibility plane shall be made, and verification of the visibility shall be performed

4.3.25 Air transportability. An air transportability report shall be submitted to the government representative at first production verification to begin the air transportability certification process. The report shall outline the steps necessary to load the tow tractor onto the specified aircraft, including diagrams, drawings, or instructions of the tow tractor preparation, loading, shoring, and tie down procedures (including engineering analysis of the tie down devices) for both the tow tractor and any removable ballasts, if applicable, along with critical dimensions, tow tractor weights, axle weights, etc.

4.3.26 Transportation data plate. The first production tow tractor shall be inspected for verification of the specified transportation data plate.

4.3.27 Identification plate. The first production tow tractor shall be inspected for verification of the specified identification plate and a rear license plate mounting bracket.

4.3.28 Lubrication data plate. The first production tow tractor shall be inspected for verification of the specified lubrication data plate.

4.3.29 Miscellaneous specifications. The first production tow tractor shall be inspected for verification of compliance with SAE specifications ARP1247C and AIR 1363A.



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4.4 Reconfiguration criteria. Failure to comply with the requirements of this CID, reference documents, or the tests shall be cause for reconfiguration and re-demonstration. Causes for reconfiguration shall include: structural failure; non-correctable misalignments; component interference; conditions presenting a safety hazard to the user or maintainer: instability during operation, spillage of fuel or coolant, overheating, and driveline vibration or shudder due to sub-optimal driveline angles.

4.5 Operational tests. If specified in the ordering data (see 6.2 i), contractor shall provide a specified number of sample units to be placed in service at representative Air Force bases. These sample units shall be used interchangeably with existing FLTTs to determine their fitness for use. The period of testing and evaluation shall be 90 days.

5. PACKAGING, PACKING, and MARKING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order (see 6.2 b).

6. NOTES.

6.1 Addresses for obtaining copies of referenced documents.

Copies of Department of Defense specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

Application for copies of Society of Automotive Engineers (SAE) documents should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

6.2 Ordering data. The purchaser shall specify the following at the time of purchase:

a. Title, number, and revision letter, and date of this document.
b. Packaging requirements (see 5.1).
c. If heavy-duty winterization is required (see 2.4).
d. When an air-system for trailer towing is required (see 2.12 and 2.15).
e. Color required (see 2.22).
f. When four-wheel drive is required (see 2.5).
g. When driver's side spotlight is required (see 2.16).
h. When Air Conditioning is required (see 2.15).
i. Operational tests if required (see 4.5).



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6.3 Subject term (key word) listing.

Aircraft parking
Bobtail
FLTT
Flight line
Ground handling
Tug

Custodians:
Air Force – 99
Army – AV

Preparing activity:
Air Force – 84

Agent
Air Force – 99

Civilian coordinating activity:
GSA/FSS

(Project 1740-0190)

FSC 1740

"NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil/ .



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